

SUMMARY

Ph.D. candidate in Biostatistics specializing in **Reinforcement Learning (RL)**, **Causal Inference** and **Large Language Models (LLMs)**. Experienced in designing and deploying statistical and machine learning models to extract insights and drive informed decision-making from complex datasets, particularly in mobile health and personalized interventions. Passionate about leveraging modern statistical and machine learning tools to create impactful products and solutions.

EDUCATION

University of Michigan <i>Ph.D. in Biostatistics</i> <i>Advisors: Dr. Zhenke Wu and Dr. Chengchun Shi</i>	Ann Arbor, MI 2020.9 - 2025.8 (<i>expected</i>)
University of Michigan <i>M.S. in Biostatistics</i>	Ann Arbor, MI 2017.9 - 2019.4
Shanghai Jiaotong University <i>B.S. in Bioinformatics</i>	Shanghai, China 2013.9 - 2017.6

SKILLS SUMMARY

Coding Language: Python (advanced), R (advanced), C/C++ (intermediate), SQL (intermediate).
Machine Learning Tools: PyTorch, TensorFlow, Hugging Face, Scikit-learn, Pandas, Scipy, Transformers.
Statistical Expertise: Reinforcement learning, Causal inference, Time series analysis, Large Language Model.

SELECTED RESEARCH EXPERIENCE

Fine-Tuning LLM for Personalized Mobile Health Interventions	2025.01 - <i>present</i>
- Processed training data from wearable devices and surveys from a six-month study with 2000+ medical interns. - Fine-tuned a pretrained GPT-2 model using RLHF (SFT and PPO) to generate tailored and diverse messages.	
Counterfactually Fair Offline RL via Sequential Data Preprocessing	2023.4 - 2024.12
- Developed a structural causal model-based algorithm for counterfactually fair policy learning in offline RL settings. - Achieved a 20%+ reduction in disparities for vulnerable subgroups on a mobile health dataset with 200+ patients.	
Multivariate Dynamic Mediation Analysis under a RL Framework	2023.4 - 2024.8
- Developed a recursion-based statistical approach for estimating multivariate dynamic mediation effects under RL. - Applied this method to a 26-week mobile health dataset of 1200+ individuals, identifying resting heart rate and sleep as key mediators of in-the-moment mobile prompts' long-term effects on reducing medical intern's depression.	
Testing Stationarity Assumption in Sequential Decision Making	2021.11 - 2023.4
- Developed a deep learning-based hypothesis test for the stationarity assumption in high-dimensional RL settings. - Established the theoretical results of the test, proving size and double robustness under bidirectional asymptotics.	

SELECTED PUBLICATIONS

*Luo, L., *Shi, C., *Wang, J., Wu, Z., & Li, L. (2025). Multivariate dynamic mediation analysis under a reinforcement learning framework. *The Annals of Statistics*, 53(1), 400-425. (*co-first authors).

Wang, J., Shi, C., Piette, J. D., Loftus, J. R., Zeng, D., & Wu, Z. (2025). Counterfactually Fair Reinforcement Learning via Sequential Data Preprocessing. *arXiv preprint* arXiv:2501.06366.

Wang, J., Shi, C., & Wu, Z. (2023). A Robust Test for the Stationarity Assumption in Sequential Decision Making. *Proceedings of the 40th International Conference on Machine Learning*, 3635536379.

Ge, L., Wang, J., Shi, C., Wu, Z., & Song, R. (2023). A Reinforcement Learning Framework for Dynamic Mediation Analysis. *Proceedings of the 40th International Conference on Machine Learning*, 1105011097.

Wang, J., Fang, Y., Frank, E., Walton, M. A., Burmeister, M., Tewari, A., Dempsey, W., NeCamp, T., Sen, S., & Wu, Z. (2023). Effectiveness of gamified team competition as mHealth intervention for medical interns: A cluster micro-randomized trial. *Npj Digital Medicine*, 6(1), 1-8.